# Guillaume **Viejo**, PhD

## Postdoctoral fellow at the Montreal Neurological Institute

linkedin.com/in/guillaume-viejo  github.com/gviejo
+1 514 581 9912  guillaume.viejo@gmail.com
1571, rue Rachel E, Montreal, Canada

## Skills

| | |
|---|---|
| Programming | **Python**, **Numpy**, **Pandas**, Scipy, Matplotlib, Numba, Matlab, Julia, C++ |
| Frameworks | **Scikit-learn**, **XGBoost**, **Tensorflow**, Reinforcement learning, Genetic algorithms, pyQt |
| Developping tools | **Git**, Sublime-text |
| OS system | Linux |
| Autres | Latex |

## Programming projects

### PYNAPPLE : A PACKAGE TO ANALYSE NEUROPHYSIOLOGICAL DATA — 2021 - NOW

peyrachelab.github.io/pynapple  pypi.org/project/pynapple/
pynapple is a light-weight python library for time series analysis dedicated to neurophysiological data.

Python | pandas | numpy | numba

### PYNACOLLADA : COLLABORATIVE PLATFORM FOR HIGH LEVEL ANALYSIS IN SYSTEM NEUROSCIENCE — 2021 - NOW

PeyracheLab/pynacollada
Shared repository for high-level data analysis with the pynapple package

Python | pandas | numpy | numba

### USING A TENSORFLOW MODEL FOR PREDICTING BRAIN ACTIVITY — 2018 - 2019

github.com/gviejo/SWR_factors
Predicting neuronal activity using a recurrent neural network built with Tensorflow.

Tensorflow | Python

### USING XGBOOST FOR DECODING BRAIN ACTIVITY — 2017 - 2018

github.com/PeyracheLab/NeuroBoostedTrees  github.com/gviejo/Prediction_XGB_head_direction
Analyzing neuronal data with gradient boosted trees (using the package XGBoost).

XGBoost | Python

### MINISCOPY : A PACKAGE TO ANALYSE CALCIUM IMAGING DATA — 2017-2018

github.com/PeyracheLab/miniscoPy
MiniscoPy is a Python 3 implementation of CNMF-E, an efficient algorithm to deconvolve calcium transients.

Python

### HIERARCHICAL LEARNING IN C++ — 2013-2014

github.com/gviejo/ANIMAT
Agent-based simulation for studying hierarchical learning during navigation using a C++ simulator

C++

## Work Experience

**Today – January 2021** | Research Associate, MONTREAL NEUROLOGICAL INSTITUTE, Canada

- Creating packages for pre-processing and analysis of data in the lab and for the scientific community
- Mentoring PhD students.
- Monitoring and testing innovative tools for data analysis

Data analysis | PhD mentoring | Scientific writing

**December 2021 – January 2016** | Post-doctoral fellow, MONTREAL NEUROLOGICAL INSTITUTE, Canada

- Collecting and analysing large scale datasets of neurophysiological activity.
- Creating packages for pre-processing and analysis of data in the lab and for the scientific community

Data analysis | Rodent surgery | Brain recordings | PhD mentoring | Scientific writing

## Scientific publications

2020 **Precise coupling of the thalamic head-direction system to hippocampal ripples**
G Viejo, A Peyrache.
*Nature Communications, 2020.*

2019 **Thalamocortical processing of the head-direction sense**
A Peyrache, AJ Duszkiewicz, G Viejo, S Angeles-Duran.
*Progress in Neurobiology, 2019*

2018 **Brain-state invariant thalamo-cortical coordination revealed by non-linear encoders**
G Viejo, T Cortier & A Peyrache
*PLoS computational biology, 2018*

**Adaptive coordination of working-memory and reinforcement learning in non-human primates performing a trial-and-error problem solving task**
G Viejo, B Girard, E Procyk, M Khamassi.
*Behavioural brain research, 2018*

2017 **A hippocampo-cerebellar centred network for the learning and execution of sequence-based navigation**
BM Babayan, A Watilliaux, G Viejo, AL Paradis, B Girard, L Rondi-Reig.
*Scientific reports, 2017*

2015 **Modeling choice and reaction time during arbitrary visuomotor learning through the coordination of adaptive working memory and reinforcement learning**
G Viejo, M Khamassi, A Brovelli, B Girard.
*Frontiers in behavioral neuroscience, 2015*

## Education

2013 - 2016 PhD in computational Neuroscience, Institute for Intelligent Systems and Robotics, **Pierre & Marie Curie University**, Paris, France
2011 - 2013 MA in Artificial Intelligence and Decision, **Pierre & Marie Curie University**, Paris, France
2009 - 2011 MA in Neuroscience, **Caen University**, Caen, France
2006 - 2009 BA in Neuroscience, **Caen University**, Caen, France

## Teaching

2021 Summer school on python and matlab packages for extracellular recordings (Systems & Circuits Neuroscience) at the Integrated Program of Neuroscience, McGill University, Montreal, Canada.

## Poster presentation

- **2021**, Society for Neuroscience (SFN). Chicago, USA.
- **2019**, Society for Neuroscience (SFN). Chicago, USA.
- **2019**, Colloque de la Société des Neurosciences Françaises. Marseille, France.
- **2018**, Society for Neuroscience (SFN). San Diego, USA.
- **2017**, Society for Neuroscience (SFN). Washington, USA.
- **2015**, Colloque de la Société des Neurosciences Françaises. Montpellier, France.
- **2014**, Twenty Third Annual Computational Neuroscience Meeting. Quebec city, Canada.
- **2014**, Fourth Symposium on Biology of Decision-Making. Paris, France

## Scientific talks

- **2021**, Subcortical organisation of spatial signals during wake and sleep, Virtual meeting of the International Development Research Centre
- **2020**, Subcortical organisation of spatial signals during wake and sleep, Virtual symposium of the McGill University's Integrated Program in Neuroscience
- **2019**, A link between dynamics and function in the anterior thalamus, Cognition and Circuits Seminar Series of the Montreal Neurological Institute